VORTEXA 3.0

12 HRS HACKATHON

POWERED BY A.I.R.S

DOMAIN & NO : E-03

PROBLEM STATEMENT TITLE :

TeachSense AI – AI Classroom Insight & Lesson Copilot

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PROBLEM STATEMENT

THE PROBLEM

Teachers can see student scores, but understanding the reasons behind recurring mistakes takes time.

1

LIMITED INSIGHT

Scores reveal performance but may not explain the underlying learning difficulty.

2

MANUAL ANALYSIS

Identifying recurring misconceptions across many responses requires time and effort.

3

GENERIC REMEDIATION

Students with different learning needs may receive the same explanation or practice.

4

DIFFICULT PROGRESS TRACKING

Comparing results before and after targeted support can require additional manual work.

How can we transform classroom assessment data into actionable teaching decisions while keeping teachers in control?

IDEA TITLE / PROPOSED SOLUTION

TEACHSENSE AI

From Student Responses to Smarter Teaching

TeachSense AI is a teacher-facing AI-assisted platform that helps educators identify likely misconception patterns, organize students by learning needs, prepare targeted remediation materials, and compare subsequent assessment performance.

1 **UPLOAD**
Ingest assessment questions, answer keys, and student responses.

2 **CLASSIFY**
Identify likely misconception patterns from the available assessment evidence.

3 **REVIEW**
Allow teachers to confirm or correct uncertain classifications.

4 **GROUP**
Organize learners according to their learning needs.

5 **REMEDIATE**
Prepare editable, group-specific instructional materials for teacher approval.

6 **COMPARE**
Examine pre- and post-remediation assessment results.

AI assists the teacher. The teacher makes the final decision.

TECHNICAL APPROACH

1 Teacher / Educator — Assessment questions · Answer key · Student responses · Review & approval actions

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2 Web Application — Next.js App Router · React + TypeScript · Dashboard navigation · Input validation · Workflow state & browser storage

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3 Assessment Processing — Data ingestion · Response normalization · Answer-key association · Missing & unmatched record detection

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4 Misconception Analysis — Deterministic rule-based classifier · Candidate labels with evidence · Confidence below 85% flagged for review

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Teacher review — Confirm / Correct / Flag uncertain → updated learning groups

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5 Learning Support — Overlapping learning-needs groups (a student can appear in several) · Remediation drafts · Edit & regenerate

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Teacher approval — Edit, regenerate, then approve before materials are used

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6 Outcome Analysis — Pre/post comparison · Group & question summaries · Sample-size caveats → actionable learning insights

Next.js · React · TypeScript · Tailwind CSS · Zod · Vitest

UI & motion: Framer Motion · GSAP · React Three Fiber · Sensei AI companion

FEASIBILITY AND VIABILITY

FEASIBLE TODAY. EXTENSIBLE TOMORROW.

TECHNICAL FEASIBILITY

- Built with Next.js, React, and TypeScript
- Deterministic classification: predictable and testable
- Structured validation flags invalid or incomplete inputs
- Modular stages: assess, analyze, group, remediate, compare

OPERATIONAL VIABILITY

- Guided teacher-facing workflow
- Teacher review and approval
- Synthetic assessment data for safe demos and testing
- Browser-based interface

FUTURE SCOPE

- Expand misconception rules across subjects
- Evaluate model-assisted classification where useful
- Support additional assessment formats
- Improve learning analytics and reporting

Working prototype · 60/60 automated tests passing · production build compiles all 19 routes (per project README)

IMPACT AND BENEFITS

TURN ASSESSMENT INTO ACTION

FOR TEACHERS

- Reduce repetitive assessment-analysis effort
- Identify recurring difficulty patterns
- Prepare targeted remediation
- Support evidence-informed instructional decisions

FOR STUDENTS

- Receive practice targeted to identified learning needs
- Benefit from differentiated explanations
- Have learning needs considered beyond an overall score

FOR INSTITUTIONS

- Gain clearer learning-progress summaries
- Identify concepts that may need further instruction
- Support more structured academic planning

ASSESS → **UNDERSTAND** → **SUPPORT** → **REASSESS**

A continuous feedback loop between assessment, insight, remediation, and subsequent evaluation.

DIFFERENTIATOR

MORE THAN A SCORE. A COMPLETE TEACHING WORKFLOW.

TeachSense AI connects assessment evidence to teacher-reviewed insights and targeted instructional action.

ASSESSMENT → INSIGHT → REVIEW → GROUPING → REMEDIATION → COMPARISON

Misconception-Focused

Goes beyond marks to identify likely patterns in student errors.

Teacher-in-the-Loop

Keeps educators involved in reviewing and correcting classifications.

Learning-Needs Grouping

Supports overlapping learning needs rather than one category per student.

Targeted Remediation

Connects identified difficulties to editable, teacher-approved materials.

Evidence-Aware Comparison

Brings before-and-after assessment insights into the same workflow.

Our differentiator is the integration of the complete assessment-to-remediation workflow with teacher oversight.

THANK YOU!

TEACHSENSE AI

Know What Your Class Needs Next.

From assessment insights to targeted teaching.

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Questions & Discussion | Team CodeSynergy

Game's Still On

